

What is 'this'?

In Java, the `this` keyword refers to the **current instance** of the class. It is most commonly used to distinguish between class attributes and parameters when they share the same name.

Variable Shadowing

When a parameter name matches a field name, the parameter "shadows" the field. Use `this` to specify you want the object's attribute.

```
public class Main {  
    int x;  
  
    public Main(int x) {  
        this.x = x; // this.x is the class field; x is the parameter  
    }  
}
```

Constructor Chaining

You can use `this()` to call another constructor within the same class. This allows for code reuse and default value management.

Rule: The `this()` call MUST be the first statement in the constructor body.

```
public Main(String name) {  
    this(2026, name); // Calls the 2-parameter constructor below  
}  
  
public Main(int year, String name) {  
    this.year = year;  
    this.name = name;  
}
```

Execution Summary

- **Clarity:** Removes ambiguity in assignments.
- **Reusability:** Use `this()` to bridge multiple constructors.
- **Current Context:** Always points to the specific object currently running the code.